ALGEBRA

github.com/danimalabares/stack

CONTENTS

1. RINGS

Theorem 1.1. *Let R be a ring and $I \subset R$ an ideal. There is a correspondence between ideals of R containing I and ideals of R/I.*

Proof. We show that there are two maps between the set of ideals of R containing I and the ideals of R/I whose compositions are the identities in the corresponding set. One map is the projection to the quotient, and the other map is taking the elements whose equivalence class lies in a given ideal.

Suppose that $\tilde{J}$ is an ideal of R/I. Define $J = \{j \in R : j + I \in \tilde{J}\}$, the set of elements that project to $\tilde{J}$. Then $I \subset J$, because $i + I = I$ is the zero of R/I which is contained in any ideal of R/I. Also notice that J is an ideal of R since for any $r \in R$, $rj + I \in \tilde{J}$ by $\tilde{J}$ being an ideal. Notice that projecting J back to R/I gives J by definition.

Conversely, we project to the quotient. Let J be any ideal of R containing I Let $\tilde{J}$ be the projection to R/I. Then $\tilde{J}$ is an ideal of R/I by J being an ideal of R. Consider the set J' of all elements whose projection lies in $\tilde{J}$.

Then $J' = J$ by definition. $\square$

This is the essential tool in proving that R/I is a field when I is a maximal ideal

Lemma 1.2. *Let R be a ring and I a maximal ideal of R. Then R/I is a field.*

Proof. It's easy to prove that a ring is a field if and only if its only ideals are $\{0\}$ and R. By Theorem 1.1, the ideals of R/I are in correspondence with ideals of R containing I, but only R is such. $\square$

2. Modules

It looks like most books (at least [**?**] and [**?**]) define a module to be an abelian group along with a certain operation with elements of the ring.

Which secretly just says that

Definition 2.1. Let R be a ring. An *(left) R-module* M is an abelian group such that there exists a (left) ring morphism

$$R \to \mathrm{End}(M).$$

Indeed, the three usual requirements correspond to:

(1) The endomorphism corresponding to $a \in R$ respect the group structure of the group M:

$$a(x+y) = ax + ay,$$

(2) The representation map is a map of rings

$$(a+b)x = ax + by, \qquad (ab)x = a(bx).$$

3. Algebras

Definition 3.1. If R is a commutative ring, then a *commutative algebra* over R is a commutative ring S together with a ring morphism $R \to S$.

But actually I was expecting that S would be defined as a ring that is also an R-module. So let us note that a morphism of rings $R \to S$ gives a representation $R \to \mathrm{End}(S)$ via left multiplication. But the other way around, given an endomorphism associated to some element in R, how do we assign an element of S so as to produce a map $R \to S$?

Now we briefly introduce finitely-generated algebras and modules. Upshot: a finitely-generated R-algebra S is such that $S \simeq R[x_1, \ldots, x_n]/I$ for some ideal $I \subset R$. Another way of saying this is that there is a surjective morphism $R[x_1, \ldots, x_n] \to S$. Finitely-presentedness is when $I = (s_1, \ldots, s_k)$.

As an aside, recall that finitely presented k-algebras with no nilpotents are in correspondence with affine algebraic varieties.

Definition 3.2. Let $R \to S$ be a ring map.

(1) We say $R \to S$ is of *finite type*, or that *S is a finite type R-algebra* if there exist an $n \in \mathbf{N}$ and a surjection of R-algebras $R[x_1, \ldots, x_n] \to S$.
(2) We say $R \to S$ is of *finite presentation* if there exist integers $n, m \in \mathbf{N}$ and polynomials $f_1, \ldots, f_m \in R[x_1, \ldots, x_n]$ and an isomorphism of R-algebras $R[x_1, \ldots, x_n]/(f_1, \ldots, f_m) \cong S$.

For modules, the definition of finitely generated R-module is that there exists a surjective morphism $R^n \twoheadrightarrow S$. So: finitely-generated algebra means elements are polynomials on the generators with coefficients in the ring, and finitely-generated module means elements are linear combinations on the generators with coefficients in the ring.

4. Localization

The general recipe for localization is that $\mathcal{O}_U = \mathcal{O}_X[f^{-1}]$ where $U = X \setminus V(f)$. That is, localizing gives the sheaf of the variety without the zero set.

Definition 4.1. Let R be a ring and $S \subset R$ a multiplicative subset of R (containing 1). The *localization* of R at S is the ring RS^{-1} (where S^{-1} is just a set with the same elements as S but with a -1 exponent) with the operations of $(rs^{-1})(r_1 s_1^{-1}) = rr_1 s^{-1} s_1^{-1}$, $rs^{-1} + r_1 s_1^{-1} = (rs_1 + r_1 s)(s^{-1} s_1^{-1})$ and the identification given by $rs^{-1} = r_1 s_1^{-1} \iff rs_1 = r_1 s$. (Check this definition if R has zero divisors!)

In a simple case, consider a ring R and $f \in R$. The *localization of R at f* is $R[t]/(ft-1) := R[f^{-1}] = R(f)$. The quotient basically says that t is the inverse of f.

Example 4.2.
- Dyadic numbers: $\mathbb{Z}[2^{-1}]$, which are fractions of integers with denominators which are powers of 2.
- Laurent polynomials: $\mathbb{C}[t, t^{-1}]$.

Lemma 4.3. *If $R[f^{-1}] = 0$ then f is nilpotent.*

Proof. If $R[f^{-1}] = 0$ then $1 \in (ft - 1)$, which means that $1 = (1 - ft)p(t)$ for some $p(t) \in k[t]$. Then... □

The localization a ring is not always a local ring. But if S is a prime ideal, then it is.

Exercise 4.4. Prove that localization of a Noetherian ring is Noetherian.

Proof. Let R be a Noetherian ring and $S \subset R$. Let $I_1 \subset I_2 \subset \dots$ be a chain of ideals in the localization $R[S^{-1}]$. Let $\pi : R \to R[S^{-1}]$ the canonical projection $r \mapsto r$.

Let I be any of the I_i. Consider $\pi^{-1}(I)$. This gives a chain of ideals in R, which must stabilize. To go back to the localization simply recall that $\pi^{-1}(I)[S^{-1}] = I$. □

5. Noetherian and Artinian rings

In this section we explain why can any affine algebraic variety be expressed as a union of irreducible affine algebraic varieties. Roughly, the proof is as follows. Hilbert's basis theorem assures that the ring of polynomials in n variables over a field is Noetherian, which guarantees that there are only finitely many prime ideals containing any given ideal. Since the radical of an ideal I is the intersection of all primes $\mathfrak{p}_i$ containing I, and the variety of the radical is the same as the variety of the ideal, we can express $V(\sqrt{I}) = V(\cap \mathfrak{p}_i) = \cup V(\mathfrak{p}_i)$.

Definition 5.1. A ring is called *Noetherian* if any increasing chain of ideals $I_1 \subset I_2 \subset \dots$ stabilizes. It is called *Artinian* if any descending chain of ideals $I_1 \supset I_2 \supset \dots$ stabilizes.

Exercise 5.2. A ring is Noetherian if and only if all its ideals are finitely generated.

Proof. First we prove that if not every ideal is finitely generated then R cannot be Noetherian. Indeed, if I is not finitely generated we construct an ascending chain by of ideals by starting with any generator of I and adding other generators one by one.

For the converse suppose that all ideals of R are finitely generated and pick an ascending chain $I_1 \subset I_2 \subset \dots$. Consider $I = \bigcup_i I_i$, which is an ideal since for any $a \in I$ there is i such that $a \in I_i$ so that $ra \in I_i$ for any $r \in R$. However, since it is finitely generated, all the generators have to be in some I_j, and thus the chain stabilizes. □

Theorem 5.3 (Hilbert basis theorem)**.** *If R is Noetherian then so is $R[x]$.*

Exercise 5.4. If A is Noetherian then so is $A[[t]]$.

Proof. Pass from an ideal in $A[[t]]$ to A by taking the coefficients of the monomials of minimal degree of each element in the ideal. Then take a finite set of generators. Then go back to $A[[t]]$ by taking the polynomials where the generators where taken. Then. . . □

Exercise 5.5. The radical of an ideal is the intersection of all primes containing it, that is,

$$\sqrt{I} = \bigcap_{\mathfrak{p} \supset I} \mathfrak{p}_i.$$

Since in a Noetherian ring there are only finitely many primes containing any ideal, we obtain the irreducible decomposition of affine algebraic varieties.

6. Tensor product

Definition 6.1.

Theorem 6.2 (Universal property of tensor product)**.** *Let M_1, M_2, M be modules over a ring R. For any bilinear map $B : M_1 \times M_2 \to M$ there exists a unique homomorphism $b : M_1 \otimes M_2 \to M$ making the following diagram commute*

$$\begin{tikzcd} M_1 \times M_2 \arrow[r, "\pi"] \arrow[dr, "B"'] & M_1 \otimes M_2 \arrow[d, "b"] \\ & M \end{tikzcd}$$

where $\pi(m_1, m_2) = m_1 \otimes m_2$.

Remark 6.3. For $R = k$ the above property shows that $\mathrm{Bil}(M_1 \times M_2, k) = (M_1 \otimes M_2)^*$. If we assume that the vector spaces are finite dimensional, we can take duals to obtain

$$M_1 \otimes M_2 = (M_1 \otimes M_2)^{**} = \mathrm{Bil}(M_1 \times M_2, k)^*,$$

which is how Whitney first defined tensor products.

Here's Misha's way to show that the universal property of tensor product implies its uniqueness. Consider the category $\mathcal{C}$ of pairs of an R-module M, and a bilinear map $M_1 \times M_2 \to M$. Then we prove that the tensor product $M_1 \otimes M_2$ is an initial object in such category, and it is easy to show that initial objects are unique.

Remark 6.4. Misha's alternative way of stating the universal property of tensor product is that there is a bijective correspondence

$$\mathrm{Hom}_R(M_1 \otimes_R M_2, M) \leftrightsquigarrow \mathrm{Bil}(M_1 \times M_2, M).$$

The key observation is that the LHS are *linear* maps from a module to M while the RHS are *bilinear* maps.

Exercise 6.5 (Currying)**.**

$$\mathrm{Hom}(M_1 \otimes_R M_2, M) = \mathrm{Hom}(M_1, \mathrm{Hom}(M_2, M))$$

Proof. This is implied by the universal property since the RHS is just $\mathrm{Bil}(M_1 \times M_2, M)$. □

Exercise 6.6. Let

$$M_1 \xrightarrow{f_1} M_2 \xrightarrow{f_2} M_3 \longrightarrow 0$$

be exact. Show that

$$0 \longrightarrow \mathrm{Hom}(M_3, N) \xrightarrow{-\circ f_2} \mathrm{Hom}(M_2, N) \xrightarrow{-\circ f_1} \mathrm{Hom}(M_1, N)$$

is exact.

Proof. Injectivity of the first arrow is easy. For the middle term pick $f \in \mathrm{Hom}_R(M_2, N)$ in the kernel of $- \circ f_1$. To show that it is in the image of $- \circ f_2$ we look for $g \in \mathrm{Hom}_R(M_3, N)$ such that $f = g \circ f_2$.

Since $M_1 \to M_2 \to M_3 \to 0$ is exact, we have an isomorphism $\bar{f}_2 : M_2/\mathrm{img} f_1 \xrightarrow{\simeq} M_3$. Since $f \circ f_1 = 0$, we see that f induces a map $\bar{f}$ on the quotient since Indeed, if $m_2 - m_2' \in \mathrm{img} f_1$, we have $0 = f(m_2 - m_2') = f(m_2) - f(m_2')$.

Then for any $m_3 \in M_3$ we have a unique class $m_2 + M_1 \in M_2/\mathrm{img} f_1$. Thus we can define $g = \bar{f} \circ \bar{f}_2^{-1}$.

Note that this choice satisfies $f = g \circ f_2$. Indeed,

$$g \circ f_2(m_2) = \bar{f} \circ \bar{f}_2^{-1}(m_3) = \bar{f} \circ (m_2 + M) = f(m_2).$$

□

Exercise 6.7. Let $R = \mathbb{Z}$. Find an example of an injective homomorphism $M_1 \to M_2$ such that $\mathrm{Hom}(M_2, N) \to \mathrm{Hom}(M_1, N)$ is not surjective.

Proof. Consider

$$\begin{tikzcd} 0 \arrow[r] & 2\mathbb{Z} \arrow[r, "i"] \arrow[d, "\varphi"'] & \mathbb{Z} \arrow[dl, dashed, "\nexists"] \\ & \mathbb{Z}/2\mathbb{Z} & \end{tikzcd}$$

where

$$\begin{aligned} \varphi : 2\mathbb{Z} &\longrightarrow \mathbb{Z}/2\mathbb{Z} \\ 2n &\longmapsto [n]. \end{aligned}$$

Then there is no extension $\bar{\varphi} : \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$ making $\varphi = \bar{\varphi} \circ i$. Indeed, note that $\varphi(2) = [1]$, while $\bar{\varphi}(2 \cdot 1) = 2\bar{\varphi}(1) = [0]$. □

The idea is that any element in the image must be divisible by any integer. This is why $\mathbb{Q}$ is an injective module according to the following definition. (A precise proof of why $\mathbb{Q}$ is injective can be found at 01D6.)

Definition 6.8. An R-module is *injective* if the functor $\mathrm{Hom}(-, M)$ is exact. That is, if the dotted arrow in the following diagram always exists:

$$\begin{tikzcd} 0 \arrow[r] & M_1 \arrow[r, hook] \arrow[d] & M_2 \arrow[dl, dashed, "\exists"] \\ & N & \end{tikzcd}$$

Exercise 6.9. Let

$$0 \longrightarrow M_1 \xrightarrow{f_1} M_2 \xrightarrow{f_2} M_3 \longrightarrow 0$$

be an exact sequence of R-modules. Prove that

$$0 \longrightarrow \mathrm{Hom}_R(N, M_1) \xrightarrow{f_1 \circ -} \mathrm{Hom}_R(N, M_2) \xrightarrow{f_2 \circ -} \mathrm{Hom}_R(N, M_3)$$

is exact

Proof. Injectivity is simple. For the middle term let $f : N \to M_2$ be in the kernel of $f_2 \circ -$, that is, $f(N)$ is killed by f_2.

We want to define a map $g : N \to M_1$ such that $f_1 \circ g = f$. But observe that the image of f already lies in $M_1 = \mathrm{Ker}\, f_2$. Thus f is already a map $f : N \to M_1$, save a more precise notation. $\square$

Exercise 6.10. Let $R = \mathbb{Z}$. Find an epimorphism (surjective homomorphism) of R-modules $M_2 \to M_3$ such that the corresponding map

$$\mathrm{Hom}_R(N, M_2) \to \mathrm{Hom}_R(N, M_3)$$

is not surjective for some R-module N.

Proof. Now consider

$$\begin{tikzcd} & \mathbb{Z} \arrow[dl, dashed, "\nexists"'] \arrow[d, "\pi"] & \\ 2\mathbb{Z} \arrow[r, "\varphi"'] & \mathbb{Z}/2\mathbb{Z} \arrow[r] & 0 \end{tikzcd}$$

for the same φ as in Exercise 6.7. Then $\pi(1) = [1]$, while any extension mapping $1 \to 2n$ gives $\varphi(2n) = 2\varphi(n) = [0]$. $\square$

Definition 6.11. An R-module N is *projective* if the functor $\mathrm{Hom}_R(N, -)$ is exact. That is, if the dotted arrow in the following diagram always exists:

$$\begin{tikzcd} & N \arrow[dl, dashed, "\exists"'] \arrow[d] & \\ M_2 \arrow[r, two heads] & M_3 \arrow[r] & 0 \end{tikzcd}$$

Exercise 6.12. A finitely generated R-module N is projective if and only if it is a direct summand of a free R-module R^n, i.e. there is a direct sum decomposition $R^n = N \oplus N'$.

Proof. Consider $R^n \simeq \bigoplus_{i=1}^n a_i R$ where N is generated by $a_1, \ldots, a_n$. Apply projectivity to obtain

$$\begin{tikzcd} & N \arrow[dl, dashed, "\exists\psi"'] \arrow[d, "\mathrm{id}"] & \\ R^n \arrow[r, "\varphi"'] & N \arrow[r] & 0 \end{tikzcd}$$

where

$$\varphi : R^n \longrightarrow N$$
$$(r_1 a_1, \ldots, r_n a_n) \longmapsto \sum r_i a_i.$$

This is a section of φ in the exact sequence

$$\begin{tikzcd} 0 \arrow[r] & \operatorname{Ker}\varphi \arrow[r] & R^n \arrow[r, "\varphi"] & N \arrow[l, bend left, "\psi"] \arrow[r] & 0 \end{tikzcd}$$

which gives decomposition

$$R^n \simeq \operatorname{Ker}\varphi \oplus \operatorname{Im}\psi.$$

More explicitly, the intersection of those modules is clearly trivial while any element of R^n may be written as

$$x = \underbrace{(x - \psi\varphi(x))}_{\in \operatorname{Ker}\varphi} + \underbrace{\psi\varphi(x)}_{\in \operatorname{Im}\psi}.$$

For the converse we start from the diagram given in the definition of projective module, add N' to every term, arriving at $R^n \simeq N \oplus N'$. We observe that free modules are projective by constructing an extension using a basis of R^n, and then back to N by taking the action of the extension restricted to N. $\square$

The upshot about projective modules is precisely that they can't be direct summands of a free module. Thus $\mathbb{Z}_2$ is not projective, since it has torsion, and so it cannot be a summand of $\mathbb{Z}^n$.

Thus, we create a non-projective module by quotienting the ring by any ideal - an element of the ideal is nilpotent, i.e. is a torsion element.

Further, it looks like projective modules are flat (confirm this!)

In fact, currying (Exercise 6.5) is the essential ingredient for the proof that tensor product is left exact. Such result follows from two lemmas.

Lemma 6.13. *Let*

$$0 \longrightarrow M_1 \longrightarrow M_2 \longrightarrow M_3 \longrightarrow 0$$

be an exact sequence of R-modules. Then for any R-modules N and N', the sequence

$$0 \longrightarrow \operatorname{Hom}_R(M_3 \otimes N', N) \longrightarrow \operatorname{Hom}_R(M_2 \otimes N', N) \longrightarrow \operatorname{Hom}_R(M_1 \otimes N', N)$$

is exact.

Proof. Apply $\operatorname{Hom}_R(-, N)$, then $\operatorname{Hom}_R(N', -)$ and then currying. $\square$

There is an (almost) converse to Exercise 6.6:

Lemma 6.14. *Let*

$$E: \qquad M_1 \xrightarrow{\mu} M_2 \xrightarrow{\rho} M_3 \longrightarrow 0$$

be a complex, i.e. $\rho\mu = 0$*, such that*

$$0 \longrightarrow \mathrm{Hom}_R(M_3, N) \xrightarrow{\rho_N} \mathrm{Hom}_R(M_2, N) \xrightarrow{\mu_N} \mathrm{Hom}_R(M_1, N)$$

is exact for any R*-module* N*. Then* E *is also exact.*

And this is what implies right-exactness of tensor product:

Theorem 6.15. *Let*

$$M_1 \longrightarrow M_2 \longrightarrow M_3 \longrightarrow 0$$

be exact. Then

$$M_1 \otimes_R M \longrightarrow M_2 \otimes_R M \longrightarrow M_3 \otimes_R M \longrightarrow 0$$

is exact.

Proof. This is just Lemma 6.13 and Lemma 6.14!!! □

Exercise 6.16. Let $R = \mathbb{C}[t]$. Find an exact sequence of R-modules

$$0 \longrightarrow M_1 \longrightarrow M_2 \longrightarrow M_3 \longrightarrow 0$$

and an R-module N such that the corresponding map

$$M_1 \otimes_R N \to M_2 \otimes_R N$$

is not injective.

Proof. Suppose the given exact sequence is just the ideal exact sequence of a point in $\mathbb{A}^1$. Then tensoring by M means thickening the point. Not having injectivity after tensoring means that tensoring the ideal by M does not give the ideal of the resulting variety. □

And from this it follows as corollary

Proposition 6.17. *Let* $I \subset R$ *be an ideal of a ring. Then* $M \otimes_R (R/I) = M/IM$.

Proof. Take the exact sequence $0 \to I \to R \to R/I$ and tensor by M. □

Exercise 6.18. Let I, I' be distinct maximal ideals in a ring R. Then $R/I \otimes_R R/I' = 0$.

Proof. They key observation is that since I and I' are maximal we must have $I + I' = M$. This means that $1 = f + f'$ for some $f \in I$ and $f' \in I'$. Then $1(a + I) \otimes (b + I') = (fa + I) \otimes (b + I') + (a + I) \otimes (fb + I') = 0$. □

Now we discuss tensor product in the category of rings over a ring.

Definition 6.19. Let A, B and C be rings, and $C \to A$ and $C \to B$ homomorphisms. Considering A and B as C-modules, we can define a product on $A \otimes_C B$ by

$$a \otimes b \cdot a' \otimes b' := aa' \otimes bb'.$$

By the universal property this indeed extends to a product.

Example 6.20.

$$\mathbb{C}[t_1, \ldots, t_k] \otimes_{\mathbb{C}} \mathbb{C}[z_1, \ldots, z_n].$$

Example 6.21. For any homomorphism $\varphi : C \to A$, the ring $A \otimes_C (C/I)$ is isomorphic to $\frac{A}{\varphi(I)A}$ by Proposition 6.17.

Recall that the spectrum of a ring $\operatorname{Spec} R$ is the corresponding algebraic variety.

Proposition 6.22. *Let $f : X \to Y$ be a morphism of affine varieties and $f^* : \mathcal{O}_Y \to \mathcal{O}_X$ the corresponding morphism. Let $y \in Y$ and $\mathfrak{m}_y$ the corresponding maximal ideal. Then*

$$\operatorname{Spec}\left(\left(\mathcal{O}_X \otimes_{\mathcal{O}_Y} \frac{\mathcal{O}_Y}{\mathfrak{m}_y}\right) \Big/ \textit{nilradical}\right) \cong f^{-1}(y).$$

Remark 6.23. Notice that we can see $y \in Y$ as a subvariety. In fact, it is, like X, a variety *over* Y. Thus this expression for $f^{-1}(y)$ matches that of the fibered product in Proposition 6.31.

Lemma 6.24. *Let A, B be finitely generated reduced $\mathbb{C}$-algebras. If $R := A \otimes_{\mathbb{C}} B$, and $N \subset R$ nilradical (the set of nilpotent elements of R). Then*

$$\operatorname{Spec}(R/N) = \operatorname{Spec}(A) \times \operatorname{Spec}(B).$$

Now we discuss...

Lemma 6.25. *Let A be a finitely-generated $\mathbb{C}$-algebra. The intersection of all the prime ideals of A is its nilradical.*

Exercise 6.26. Let A and B be finitely generated $\mathbb{C}$-algebras, $B \to A$ a homomorphism and $\mathfrak{m} \subset B$ a maximal ideal. Then $A \otimes_B (B/\mathfrak{m})$ can contain nilpotents.

Proof. The geometric idea is to project a parabola to an axis. At the point of tangency we obtain a fat point, whose algebra is not reduced. □

However, over $\mathbb{C}$ the situation is different:

Theorem 6.27. *Let A, B be finitely generated reduced $\mathbb{C}$-algebras. Then $A \otimes_{\mathbb{C}} B$ is reduced.*

It follows that

Theorem 6.28.

$$\operatorname{Spec}(A) \times \operatorname{Spec}(B) = \operatorname{Spec}(A \otimes_{\mathbb{C}} B).$$

Now we can take preimages!

Lemma 6.29. *Let $f : X \to Y$ be a morphism of affine algebraic varieties, $f^* : \mathcal{O}_Y \to \mathcal{O}_X$ the corresponding ring homomorphism, $Z \subset Y$ a subvariety and I_Z its ideal. Denote by R_1 the quotient ring*

$$(\mathcal{O}_X \otimes_{\mathcal{O}_Y} (\mathcal{O}_Y/IZ)/\textit{nilradical} = (\mathcal{O}_X/f^*(I_Z))/\textit{nilradical}.$$

Then

$$\operatorname{Spec}(R_1) = f^{-1}(Z).$$

Proof. Analogous to preimage of a point. □

The slogan is: adding polynomials to an ideal is the same as taking tensor product. This operation is given by Exercise 6.18.

We can also consider the diagonal of a product:

Lemma 6.30. *Let M be an affine algebraic variety, its diagonal $\Delta \subset M \times M$ and $I \subset \mathcal{O}_M \otimes_{\mathbb{C}} \mathcal{O}_M$ the ideal generated by $r \otimes 1 - 1 \otimes r$ for all $r \in \mathcal{O}_M$. Then*

$$\mathcal{O}_M \otimes_{\mathbb{C}} \mathcal{O}_M / I.$$

And now we introduce fibered product!

Proposition 6.31. *Let $X \to M$ and $Y \to M$ be morphisms of algebraic varieties, $R = \mathcal{O}_X \otimes_{\mathcal{O}_M} \mathcal{O}_Y$, and R_1 the quotient of R by its nilradical. Then*

$$\operatorname{Spec}(R_1) = X \times_M Y.$$

Proof.

$$X \times Y \xrightarrow{f \times g} Z \times Z,$$

and compute $(f \times g)^{-1}(\Delta)$. By Lemma [**?**] we must have

$$A \otimes_{\mathbb{C}} B \big/ (r \otimes 1 - 1 \otimes r) = A \otimes_C B$$

where $A = \mathcal{O}_X$, $B = \mathcal{O}_Y$ and $C = \mathcal{O}_Z$. □

We shall not explain why, but coproduct in the category of reduced finitely generated algebras is direct sum.

7. Artinian algebras

The aim of this section is to prove that a commutative algebra over a field which is finite-dimensional as a vector space is a direct sum of fields, namely Theorem **??**. The key idea is that in a finite-dimensional algebra we can use Chinese Reminder Theorem applied to the irreducible decomposition of the minimal polynomial of a non-invertible element to construct an idempotent element. This equivalent to the claim that if there are no idempotents the algebra is a field. Then the decomposition into fields is just a decomposition of unity into indecomposable idempotents, each of which gives a summand eR with no idempotents, i.e. a field.

This decomposition is used to prove that the preimage of a point by a finite map of affine algebraic varieties is a finite set of points, Theorem 8.3. Moreover, it also used quite a lot in the development of Galois theory in Section 10.

In this section algebras are not assumed to have unity.

Definition 7.1. The *minimal polynomial* of an element v in a finitely-generated k-algebra R a monic polynomial $P \in R[t]$ of smallest possible degree having v as a root.

Lemma 7.2. *The subalgebra generated by v is isomorphic to $k[t]/(P)$.*

Proof. Consider the map $k[t] \to Rv$, whose kernel is (P) since P divides any polynomial vanishing in v. □

Example 7.3.

$$\frac{\mathbb{R}[t]}{t^2 + 1} \simeq \mathbb{C}$$

while

$$\frac{\mathbb{R}[t]}{t^2 = 1} \simeq \mathbb{R} \oplus \mathbb{R}.$$

Recall that $k[t]$ is a UFD. Also recall that P is an irreducible element of $k[t]$ if and only if $k[t]/(P)$ is a field. This follows since $k[t]/(P)$ cannot have zerodivisors if P is prime, which is equivalent to irreducible in UFDs. These extensions are called *primitive*. [They are extensions of k obtained by adding a root of P.]

Definition 7.4. A commutative, associative k-algebra R not necessarily with unit is called an *Artinian algebra* if it is a finite-dimensional vector space over k. It is called *semisimple* if it has no nonzero nilpotents.

A commutative algebra with unity over a field is called an *Artinian ring* if it is finite-dimensional as a vector space over k.

We will show that any semisimple Artinian algebra is a direct sum (direct sum in the category of algebras is the direct sum of the rings equipped with natural operations) of fields, and this decomposition is uniquely defined.

An idempotent element e in R, i.e. $e^2 = e$, gives us a canonical decomposition

$$eR \oplus (1-e)R$$

since $1-e$ is also idempotent.

Exercise 7.5. Let $v \in R$ be an element of a finite-dimensional algebra R over k. Let $k[v]$ be the subalgebra generated by $1, v, v^2, \ldots$. Prove that $P(v) = 0$ for a unique monic polynomial P of degree the dimension of $k[v]$.

Proof. Division algorithm. □

The polynomial above is called the *minimal polynomial* of v.

Exercise 7.6. Let $v \in R$ be an element of an Artinian ring over k, and $P(t)$ its minimal polynomial. Prove that $k[v]$ is isomorphic to $k[t]/(P)$.

Proof. Evaluation map. □

Exercise 7.7. Let $R = k[t]/P$ where $P \in k[t]$ is a polynomial decomposing as a product $P = P_1 \ldots P_n$ of coprime polynomials. Prove that there exists an isomorphism $R \to \bigoplus_i k[t]/P_i$.

Proof. The natural map $[F]_P \mapsto ([F]_{P_1}, \ldots, [F]_{P_n})$ is well defined. It is surjective and injective by the Chinese reminder theorem. If $[F]_P = [t]_P$, i.e. $F - t = PQ = P_1 \ldots P_n Q$, it's immediate that $[F]_{P_i} = [t]_{P_i}$ for all i. □

Exercise 7.8. Let R be a semisimple Artinian ring with all idempotents equal to 1 or 0. Prove it is a field.

Proof. Suppose R is not a field, thus there is an element x which is not invertible. Consider the algebra $k[x]$ generated by x. By a Exercise **??** we know that $k[x] \simeq k[t]/P$ where P is the minimal polynomial of x. Now decompose P in its unique irreducible factorization. Since there are no nilpotents in R, there are no nilpotents in $k[x] \simeq k[t]/P$, which implies that the irreducible factorization of P is square-free, which means that the factors of P are coprime and we can apply the Chinese Reminder theorem. That is, by Exercise 7.7, we know that $k[t]/P$ is a direct sum of fields $\bigoplus k[t]/P_i$. There is an idempotent element here: $(1, 0, \ldots, 0)$. □

Two idempotents $e_1, e_2 \in R$ are called *orthogonal* if $e_1 e_2 = 0$. If e_2, e_3 are orthogonal idempotents, then $e_1 = e_2 + e_3$ is also an idempotent satisfying $e_2, e_3 \in e_1 R$ and $e_1 R = r_2 R \oplus e_3 R$. In characteristic $\neq 2$, for any three idempotents e_1, e_2, e_3, if $e_1 =$

$e_2 + e_3$ then e_2, e_3 are orthogonal. An idempotent $e \in R$ is called *indecomposable* if there are no non-zero orthogonal idempotents such that $e = e_2 + e_2$.

Exercise 7.9. Let R be a semismple Artinian algebra, and $e \in R$ a non-decomposable idempotent. Then eR is a field.

Proof. Using Exercise **??**. If there is an idempotent er we can decompose e as a sum of orthogonal idempotents, $e = (e - er) + er$. □

The next exercise show that there is an indecomposable-orthogonal- decomposition of unity in an Artinian ring:

Exercise 7.10. Let R be a semisimple Artinian ring over a field k, char $k \neq 2$. Prove that 1 can be decomposed to a sum of indecomposable orthogonal idempotents and such decomposition is unique.

Proof. If 1 is indecomposable we are done. If it is decomposable, say $1 = e_1 + e_2$ then we can write $R = e_1 R \oplus e_2 R$. This implies that $e_1 R$ and $e_2 R$ are vector subspaces of R of strictly less dimension, and thus we can repeat the process and finish in finitely many steps. For uniqueness we write $1 = e_1 + \ldots + e_k = e'_1 + \ldots + e'_{k'}$. Multiplying by e_1 we obtain $e_1 = e_1 e'_1 + \ldots + e_1 e'_{k'}$ which, if the decompositions do not coincide, is a decomposition of e_1 into orthogonal idempotents, a contradiction. □

Exercise 7.11. Let R be an Artinian ring over a field k, char $k \neq 2$. Then R is a direct sum of fields and this decomposition is unique.

Proof. This is just Exercises 7.10 and 7.8. □

Exercise 7.12. Does it work in characteristic 2?

Proof. ChatGPT: should work but not this technique. Me: although we can decompose $R = eR \oplus (1 - e)R$, we cannot show this is a field unless we apply Chinese Reminder Theroem! Perhaps this was a mistake in the lectures, or else the whole orthogonality business is useless. □

Exercise 7.13. Let R be an Artinian ring over a field k, char $k \neq 2$ and $1 = e_1 + \ldots + e_n$ a decomposition of 1 to a sum of indecomposable orthogonal ideompotents. Prove that R has precisely n prime ideals.

Proof. My idea is that the given decomposition of unity gives a decomposition of the algebra in fields, which we may just think as an affine algebraic variety which is just n points. Any prime ideal in the algebra is an irreducible subvariety of the n points, which in turn can only be a point, which is already one of the obvious n prime ideals. □

The following result was proved in lectures:

Theorem 7.14. *An Artinian algebra without nilpotents contains an idempotent.*

The proof of the theorem constructs a maximal ideal $I \subset A$ which contains a unit, and thus is a field I.

Thus we obtain

Theorem 7.15. *An Artinian algebra with unity with no nilpotents is a direct sum of fields.*

Proof. We find an idempotent element e, so that we can decompose

$$A = eA \oplus (1-e)A$$

where e is a unity on the first factor and so is $1-e$ on the second factor, thus every component is a field [I think this might be wrong!!! In the home assignment we needed Chinese Reminder Theorem to get the fieldness; see Exercise 7.8] We repeat this process until we exhaust the dimension of A. $\square$

We can also prove this theorem for algebras without unity. We consider the kernel of the map $u_0 : a \mapsto au$ and put

$$A = uA \oplus \operatorname{Ker} u_0 A.$$

Lemma 7.16. *Any decomposition of an algebra into a direct sum of fields is unique up to permutation of the factors.*

8. Finite morphisms

Remark 8.1. If M is a finitely generated R-module and $R \to R'$ is a ring morphism, then $M \otimes_R R'$ is a finitely-generated R'-module generated by the same generators of M.

Definition 8.2. A *finite morphism* of affine varieties is called *finite* if the $\mathcal{O}_X$ is a finitely generated $\mathcal{O}_Y$-module. In this case, $\mathcal{O}_X$ is called an *integral extension* of $\mathcal{O}_Y$.

Theorem 8.3. *Let $f : X \to Y$ be a finite morphism of affine varieties. Then for any point $y \in Y$, the preimage $f^{-1}(y)$ is finite.*

Proof. Let R be the $\mathcal{O}_Y$-module $R := \mathcal{O}_X \otimes_{\mathcal{O}_Y} \mathcal{O}_Y/\mathfrak{m}_y$, which is the coordinate ring of $f^{-1}(y)$ by Proposition 6.22 or Lemma 6.29.

Further, it is finitely generated because $\mathcal{O}_X$ is finitely generated over $\mathcal{O}_Y$ (Remark 8.1).

Now observe that R is an Artinian algebra: elements of $\mathcal{O}_X$ are finite $\mathcal{O}_Y$-linear combinations of certain elements $f_1, \dots, f_n \in \mathcal{O}_X$, but in R we have tensored by $\mathcal{O}_Y/\mathfrak{m}_y$, so by definition of tensor product the $\mathcal{O}_Y$-coefficients pass to the $\mathcal{O}_Y/\mathfrak{m}_y$ slot, where they collapse to complex numbers; thus R is a finite-dimensional vector space over $\mathbb{C}$. By Theorem 7.15, $\operatorname{Spec}(R)$ is a finite set! $\square$

9. Trace form

It's not yet clear why do we need the trace form for. The main theorem in this section states that a semisimple Artinian algebra has nondegenerate trace form. This is reminiscent of Cartan's theorem which says that a finite-dimensional Lie algebra is semisimple in the sense that can be decomposed as a direct sum of simple Lie algebras (i.e. without nontrivial proper ideals) if and only if the Killing form (i.e. trace of adjoint operators) is nondegenerate.

Definition 9.1. Let R be a k-algebra, and $g : R \times R \to k$ a symmetric k-bilinear form on R. g is called *invariant* if

$$g(x, yz) = g(xy, z).$$

If R has a unity we have

$$g(x, y) = g(xy, 1),$$

so that in fact g is determined by a linear functional $a \mapsto g(a, 1)$.

Example 9.2. In the ring $\mathbb{C}[x,y]/(x^{n+1},y^{n+1})$ the form $\varepsilon(xy)=a_n n$ is nondegenerate.

Definition 9.3. Let R be an Artinian ring over k. The *trace form* is the bilinear form $a,b\mapsto \operatorname{tr}(ab)$ where $\operatorname{tr}(ab)$ is the trace of the left-multiplication by ab endomorphism, i.e. $L_{ab}\in \operatorname{End}_k R$, $x\mapsto abx$.

Exercise 9.4. Let $[K:k]$ be a finite field extension in characteristic 0. Prove that the trace form is always non-degenerate.

Proof. □

Definition 9.5. A field extension is *separable* if the trace form is nonzero.

Theorem 9.6. *Let R be an Artinian algebra over k with nondegenerate trace form. Then R is semisimple.*

Lemma 9.7. *The trace in the tensor product is the tensor product in of the traces.*

10. Field extensions

In this section we aim to prove: let $[k:k]$ be a finite extension over with $\operatorname{char}k=0$. then there exists an element $l\in k$ generating k.

We start with a motivating introduction. Field of rational functions on a variety, take its covering, remove its ramification locus, obtain a field extension. Study automorphisms of the field extensions, which is the same as studying automorphisms of the cover. This is Galois theory. Decomposition of a direct sum of fields shall correspond to the disjoint union of varieties.

A *field extension* of the field k is another field K which cointains k. This is denoted $[K:k]$.

$[K:k]$ is *algebraic* if K is finite-dimensional over k. Otherwise it is *trascendental.*

An element $x\in K$ is *algebraic* if it is a root of a nonzero polynomial with coefficients in k, and otherwise it is *trascendental.*

Exercise 10.1. A field extension is algebraic if and only if all its elements are algebraic.

A field is called *algebraically closed* is all algebraic extensions are trivial. This is equivalent to the field containing the roots of all polynomials.

Lemma 10.2. *A field k is algebraically closed if and only if any polynomial $P\in k[t]$ is a product of linear factors.*

Proof. If P does not decompose, there is a polynomial of degree >1 diving P, and the quotient $k[t]$/such polynomial is an algebraic field extension. □

There is a good notion of algebraic closure of a field obtained by adding roots and applying Zorn's lemma to ensure this process finishes. And it is unique.

Theorem 10.3. *Let $[\bar{k}:k]$ be the algebraic closure of k, and $[K:k]$ a separable finite extension. Then*

$$K\otimes_k \bar{k}=\bigoplus \bar{k}.$$

Recall that by Galois theory we can see that there only finitely many field extensions. Namely Galois theory says that field extensions are in correspondence with subgroups of the Galois group, which is finite. But this a simpler case:

Lemma 10.4. *Let k be a field, and $A = \bigoplus_{i=1}^n k$. Then A contains only finitely many different k-algebras.*

Proof. □

Now we can prove (without Galois theory):

Theorem 10.5. *let $[k : k]$ be a finite extension over with char$k = 0$. then there exists an element $l \in k$ generating k.*

Proof. Using finiteness of the contained algebras... And the following fun claim: if V_i are positive codimension subspaces of k^n, then $\bigcup V_i \neq k^n$. □

Definition 10.6. Let $[K : k]$ be a finite extension. It is called a *Galois extension* if the algebra $K \otimes_k K$ is isomorphic to a direct sum of several copies of K.

Example 10.7. Let $K = k[t]/(P)$ be a primitive, separable extension with deg$P = n$. Suppose that P has n roots in K. Then $[K : k]$ is Galois. Indeed, this is the Chinese reminder exercise once again:

$$K \otimes_k K = \frac{K[t]}{(P)} = \bigoplus_i \frac{K[t]}{(t - a_i)}.$$

Exercise 10.8. $K = k[t]/P$. $[K : k]$ is Galois if and only if P has n roots in $K[t]$.

Definition 10.9. Let $P \in k[t]$ be an irreducible polynomial. *Splitting field* is the minimal extension such that P factors into linear factors.

Equivalently, pick $P \in k[t]$ of degree n without multiple roots, and k of characteristic 0. Decompose P into irreducible factors and construct finite extensions $K_1 \subset \ldots \subset K_\ell$ by adding the roots of each factor. This process ends (prove this) and the resulting field is the *splitting field*.

Splitting field is Galois by Exercise 10.8. We can think of the splitting field as embedded in $\bar{k}$. There is not a unique embedding but all embeddings are isomorphic.

Definition 10.10. If $[K : k]$ is a Galois extension, $\mathrm{Aut}_k K$ is called *Galois group*.

Note that K^* acts on $K \otimes_k K$ by multiplication on the right and on the left.

Lemma 10.11. *Let $[K : k]$ be a Galois extension. There is a bijection between the following sets:*

(1) *$Aut_k K$.*
(2) *Prime ideals in $K \otimes_k K$.*
(3) *Homomorphisms $K \otimes_k K \to K$ which are linear with respect to the left action of K.*

Proof. ($2 \implies 3$) is simple. □

Lemma 10.12. *Let $[K : k]$ be a Galois extension, and $K \otimes_k K = \bigoplus_{\nu \in Aut_k K} K_\nu$ the decomposition of $K \otimes_k K$ enumerated by elements of $Aut_k K$. Then the left μ_l and right μ_r actions of K^* on $K \otimes_k K$ are such that $\mu_l(a)e_\nu = \mu_r(\nu(a))e_\nu$.*

Lemma 10.13. *Let $[K : k]$ be a Galois extension, and $a \in K$ a $Aut_k K$-invariant element. Then $a \in k$.*

Lemma 10.14. *Let $[K : k]$ be a Galois extension, and K' an intermediate field (which, as we will see in the proof, will also be Galois) Then*

$$K' = K^{G'}$$

where $G' \subset Aut_{K'}K$ is the group of K'-linear automorphisms of K and $K^{G'}$ is the space of G'-invariants.

Theorem 10.15 (Main theorem of Galois theory)**.** *Let $[K : k]$ be a Galois extension. The subgroups $H \subset Aut_k K$ are in bijective correspondence with intermediate subfields $k \subset K^H \subset K$.*

11. Normal varieties

Recall that $\mathrm{Spec}(R)$ is the set of all prime ideals of the ring R. We introduce a topology whose closed sets are the prime ideals contained in a given ideal.

Definition 11.1. A morphism of algebraic varieties $f : X \to Y$ is called *dominant* if its image is dense, i.e. Y is the Zariski closure of $f(X)$.

Proposition 11.2. *A morphism of varieties $f : X \to Y$ is dominant if and only if $f^* : \mathcal{O}_Y \to \mathcal{O}_X$ is injective.*

Proof. Suppose f^* is not injective and consider its kernel $\mathrm{Ker}\, f^*$. Then the image of f is in the zero set of the kernel, $V(\mathrm{Ker}\, f^*)$. Here we use that points of a variety are homomorphisms from the variety to $\mathbb{C}$, which follows from the Nullstellensatz... Dominantness means that the image is not contained in the zero set of an ideal. □

Lemma 11.3. *If $f : X \to Y$ is dominant and X is irreducible, then Y is irreducible. Moreover, $f^* : \mathcal{O}_Y \to \mathcal{O}_X$ can be extended to the fields of fractions $k(Y) \to k(X)$.*

Proof. Since f is dominant, f^* is injective, i.e. an embedding, irreducibility transforms to not having zerodivisors. Descending to the field of fraction also follows from not having zerodivisors. □

Remark 11.4. All of a sudden we consider a divisor on the affine variety X given by $f \in \mathcal{O}_X$. Then $X \setminus D$ is also an affine variety given by the ideal $I + \langle ft - 1\rangle \subset \mathbb{C}[x_1, \dots, x_n, t]$. Recall that the localization of a ring by an element is $\mathbb{C}/\langle 1 - ft\rangle$, where t is the inverse of f.

We also define *birational morphism* if the induced map on fraction fields is an isomorphism, and further show:

Proposition 11.5. *Let $f : X \to Y$ be a birational morphism. Then there exists a divisor $Z \subset Y$ such that $f : (X \setminus f^{-1}(Z)) \to Y \setminus Z$ is an isomorphism.*

Now we introduce normal varieties.

Here's some notes from the past that may pop up in memory: The upshot is: let $\varphi : R \to S$ be a ring map. An element $s \in S$ is integral over R if there is a monic polynomial with coefficients in R that gives zero when you put s instead of the variable.

Definition 11.6. Let $A \subset B$ be rings. An element $b \in B$ is called *integral over A* if the subring $A[b] = A \cdot \left\langle 1, b, b^2, \dots \right\rangle$ is generated by b and A, is finitely generated as an A-module.

Note that if A is finitely generated, $A\left\langle 1, b, b^2, \dots \right\rangle$ is obviously finitely generated as a ring, but not as a module, which is what we care about.

Lemma 11.7. *An element $x \in B$ is integral over $A \subset B$ if and only if the chain of submodules*

$$A \subset A \cdot \langle 1, x \rangle \subset A \cdot \left\langle 1, x, x^2 \right\rangle \subset \dots$$

terminates. Equivalently, x is the root of a monic polynomial with coefficients in A.

Proof. The chain statement is almost tautological (recall that being noetherian is the same as...). As for the second, express x^n as an element of the finitely-generated module. For the converse of the last equivalence we use division algorithm on the minimal polynomial to kill any residue, and thus the minimal polynomial divides any other polynomial in the algebra generated by the element. □

Lemma 11.8. *In Noetherian rings, sums and products of integral elements is integral.*

Definition 11.9. An irreducible affine algebraic variety is called *normal* if its coordinate ring is integrally closed, that is, it coincides with its its integral closure, which in turn is the set of all elements that are integral over its field of fractions.

Recall from [Har77, Exercise 3.17, Chapter 1] that for affine varieties this is equivalent to all the local rings are all integrally closed.

Proposition 11.10. *An affine variety X is normal if and only if for any finite, birational morphism $Y \to X$ is an isomorphism.*

Proof. They have the same fraction field, and $\mathcal{O}_X \subset \mathcal{O}_Y$ by dominant (birational maps are dominant), which ways that $\mathcal{O}_Y$ is finite over $\mathcal{O}_X$, which says that elements of $\mathcal{O}_Y$ are integral. Since X is normal then the elements of $\mathcal{O}_Y$ belong to $\mathcal{O}_X$. □

Proposition 11.11. *A UFD is integrally closed.*

Proof. Let u, v be elements of the UFD A and suppose the $u/v \in k(A)$ is integral, i.e. there is a monic polynomial $P \in A[t]$ of degree n killing u/v. Then u^n is divisible by v. Write $v = p_1 \dots p_k$. Then p_i divides u^n, which implies that it divides u. But then p_i divides both u and v, which we may take to be coprime, making p_i a unit for all i. □

As a corollary, we see that taking product of an affine variety, whose coordinate ring is an UFD, with affine space, $X \times \mathbb{C}^n$ preserves normality since the resulting coordinate ring is $\mathcal{O}_X[x_1, \dots, x_n] = \mathcal{O}_X \otimes \mathbb{C}[x_1, \dots, x_n]$.

12. Nakayama's lemma

We would like to have an analaogue of the inverse function theorem, i.e. a function that is surjective on the tangent bundle is locally invertible. However, this is not true. Consider B a locally free $\mathcal{O}_M$-module.

The idea is to find local trivializations of projective modules, just as we have local trivializations of vector bundles.

Nakayama's lemma is sometimes also called Krull lemma. We motivate with the following question: for an ideal $I \subset A$, when is $\cap I^n = 0$? Answer: when

$$I \cdot \bigcap I^n = \bigcap I^n.$$

Definition 12.1. Let A be a domain an M be an A-module. A *torsion* is the set of all $m \in M$ such that there exists $0 \neq a \in A$ such that $am = 0$.

Lemma 12.2 (Nakayama). *Let A be a domain and M a finitely-generated A-module. If $M = aM$ for some $a \in A$ then $M = 0$.*

Recall that we call a module *cyclic* when it is generated by a single element, i.e. $M = \langle x \rangle$. This gives a surjective map $A \to M$, $a \mapsto ax$ and so $M = A/\operatorname{Ker}$. So cyclcic modules are quotients of the ring by an ideal.

Lemma 12.3 (Nakayama). *M finitely generated A-module $\mathfrak{a} \subset A$ an ideal, $\mathfrak{a}M \subset M$. Then there exists $a \in \mathfrak{a}$ such that*

$$(1 - a)M = M.$$

Proof. First prove it for cyclic M. ** Rest of the proof pending. $\square$

We won't prove the following theorem:

Theorem 12.4 (Cayley-Hamilton). *Let M be a finitely-generated A-module. An A-linear map $\varphi : M \to M$ can be represented by a matrix, which has a characteristic polynomial P. Then $P(A) = 0$.*

The idea is to use this theorem to express the identity of M as a matrix with coefficients in A:

$$\begin{tikzcd} A^n \arrow[r, "\tilde{\varphi}"] \arrow[d, "\pi"'] & A^n \arrow[d, "\pi"] \\ M \arrow[r, "\varphi = \mathrm{Id}_M"'] & M \end{tikzcd}$$

In fact, this almost concludes the proof of Nakayama's lemma from Atiyah-Macdonald.

Now we turn to applications. Recall that localization by a prime ideal is a local ring. This says that elements outside the maximal ideal are invertible. The converse is also true and is used along with Nakayama's lemma to prove the following result. The intuition behind this is that if a map of modules induces a surjection on local quotients, then it is invertible locally (inverse function thorem!).

Lemma 12.5. *Let $(A, \mathfrak{m})$ be a local ring and M_1, M_2 be finitely generated A-modules. Let $\varphi : M_1 \to M_2$ and suppose it induces a surjective map $M_1/\mathfrak{m}M_1 \to M_2/\mathfrak{m}M_2$. Then φ is surjective.*

This easily implies

Lemma 12.6. *Let M be a finitely generated A-module, A local. Let $x_1, \dots, x_n \in M$ be generators of M mod $\mathfrak{m}$. Then $x_1, \dots, x_n$ generate M.*

Theorem 12.7. *A finite dominant map is surjective.*

Proof. $A = \mathcal{O}_Y$, $B = \mathcal{O}_X$, $\mathfrak{m}_y \in A$ maximal ideal of y. We need dominance to suppose that $B \subset A$.

Then $\mathfrak{m}_y B \neq B$ by Nakayama's lemma.

Now consider $f^{-1}(y) = \operatorname{Spec}(B \otimes_A A/\mathfrak{m}_y) = \operatorname{Spec}(B/\mathfrak{m}_y B) \neq 0$. $\square$

Recall that projective modules are summands of free modules. Vector bundles over affine varieties are the same as projective modules over the rings of functions! That's a result by Serre, but we won't prove it here.

Theorem 12.8. *A finitely generated projective module over a local ring is free.*

Proof. Consider a map

$$\begin{aligned} A^n &\longrightarrow P \\ (a_1, \ldots, a_n) &\longmapsto \sum a_i p_i. \end{aligned}$$

It is surjective and thus has a section since P is projective.

In writing $A^n = P \oplus Q$ we kill Q when taking quotient by the maximal ideal since $\mathfrak{m}Q = Q$ by Nakayama. $\square$

As a final application consider the Groethendick group $K_0(R)$ where the subindex 0 denotes that we consider only exact sequences of projective modules.

Proposition 12.9. *Let $(A, \mathfrak{m})$ be a local ring. Then $K_0(A) = \mathbb{Z}$.*

Proof. Map $P \in K_0(Q)$ to $\dim_{A/\mathfrak{m}} P/\mathfrak{m}P$. Since projective modules are free, this is a bijection. $\square$

13. Gröbner theory

So far here's some rather unformatted notes from Lin's talk at Commutative Algebra school in Recife 2025.

Would SAGBI bases be related to this?

square free monomial ideals preserve regularity upon Grobner deformation i.e. initial ideal!

Constantini-Fouli-Goel-Lin-Lindo-Liske-Mostafazadehfard, 2025.

https://sites.psu.edu/kul20/

How is Lin using the Reese algebra here? Is it ture that the initial ideal of the Gröbner deformation coincides with the initial ideal "of the Resse algebra"?

References

[Har77] Robin Hartshorne, *Algebraic Geometry*, Graduate Texts in Mathematics, vol. 52, Springer, 1977.